

A popular hashing algorithm is MD5 (Message-Digest algorithm 5) which always produces a hash of 128 bits. So for any input, whether is a 3-character string, or if it is a video file of a few gigabytes, the hash which MD5 will produce will be just 128-bit long. Another popular hashing function SHA-1 (Secure Hash Algorithm) produces 160-bit hashes. SHA1 has been superseded by SHA-2, which has four function that produce hashes of 224bit, 256bit, 384bit, or 512bit.

MD5 for example will create hashes such as the following:

```
"password"      : "5f4dcc3b5aa765d61d8327deb882cf99"  
"iddqd"         : "73bcaaa458bff0d27989ed331b68b64d"  
"iddqD"         : "93cbd5c967c1d2882aadae3950ea22be"
```

The following are examples of SHA-1 hashes:

```
"password"      : "5baa61e4c9b93f3f0682250b6cf8331b7ee68fd8"  
"iddqd"         : "67de5036a631cad20dbfe5600d6b0060fa0ce03a"  
"iddqD"         : "07735670770727d25c67d1879d5733724ec4e4ca"
```

As you can see, even a small change in the input string (from "iddqd" to "iddqD") results in a completely different hash.

Hashes can be used for other purposes than just securing passwords. You might be familiar with their usage by download sites for verifying the integrity of a download. A download site may provide you with an MD5 / SHA-1 hash for the file so that once you have downloaded the file you can verify that the download is complete and intact. Any difference in the hash of the file means that the download is possibly corrupted or malicious. If the file can be downloaded from multiple mirrors, this also ensures that external websites do not provide malicious content with the same name.

Torrent files use SHA-1 hashes of each "piece" of the content you are downloading. In torrents, the download content is divided into sections / pieces of fixed size from 16kB to 4096kB, and each piece of the download is hashed separately. This way if you are downloading a 10GB file – or ten 1GB files – in case of an error / shutdown / crash, only the corrupted piece of the file needs to be re-downloaded.

Hashes have many security related usages, hence their importance to hackers black-hat and white-hat alike. Hashes find uses in message authentication, and digital signatures as well. Hashes are equivalent to fingerprints of digital data, and understanding them can help a lot in hacking and preventing hacking.

In the next chapter, we explore the weaknesses in hashes, how they can be exploited and possible ways to secure them.